Product suggestions and customer groups: what we found

To: Sarah Okafor, Head of E-Commerce
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Hi Sarah, the short version is that both things work. Our suggestions beat the best sellers on purchases the model had never seen, the nightly job runs in under a second, and your customers fall into three groups marketing can name. Here's the detail, including where we'd be careful.

Do the suggestions beat the best sellers?

Yes. We held back the last two months of orders (9 October to 9 December 2011) and built everything from the orders before that. Then, for each returning customer who bought something new in those two months, we checked whether at least one of the new products was in the top 10 we would have suggested. It was for 38% of the 1,913 customers, against 27% when you just show the ten best sellers. That's an 11 point gap (somewhere between 9 and 14), so it isn't a fluke.

Which method, and why

We used SVD. Picture a huge grid of customers against products (5,471 by 3,686) that is 98% empty. SVD squeezes it into 100 "taste" dimensions, so products that similar shops buy end up close together. We also tried PCA, which is the same idea after subtracting the average customer first. It scored the same (38.5% against 38.3%), so we kept SVD, which works on the grid as it is.

What we lose by squeezing

About half the detail. The 100 dimensions keep roughly 47% of the original table, and what drops out is mostly one-off buying. We also learned something the hard way: our first scoring ignored how popular each product is and did worse than the best sellers (23% against 27%). The final version keeps popularity as well as each customer's own history, and that's the one behind the 38%. Some people and products can't get suggestions at all. New customers and products with fewer than 10 buyers were involved in about 3 in 10 of the new purchases we tested.

How big and how fast

The full grid is 154 MB and the squeezed version is 7 MB, about 22 times smaller. The whole nightly job (refit the model, find 10 similar products for every product, make 10 suggestions for every customer) takes under a second on a laptop. One honest caveat: stored without its empty cells, the grid is only about 5 MB, so against that the squeezing saves little. What you really get is better suggestions and quick scoring. We didn't time the old prototype, so we can't say how much faster we are than it.

Customer groups for marketing

- Everyday small shops: 83% of customers (4,543), about 47 different products each, 92% in the UK.
- Big buyers, home decor and gifts: 10% (538), about 230 products each. Hearts, hanging T-light holders, wicker, French kitchen signs. 98% in the UK.
- Big buyers, party, baking and kids: 7% (390), about 200 products each. Lunch bags, cake cases, backpacks. 29% are outside the UK, mostly Germany and France.

The edges between groups are soft, and the small shops group is big enough to split by theme later.

What could go wrong

- We tested one period, and it was the run up to Christmas. A quiet period might look different.
- 23% of order lines have no customer number, so guest buyers are invisible to us.
- The 38% is for personal top-10 lists. The product page box uses a similar-products list instead. It looked sensible (a Regency cakestand sits next to Regency teacups) but it needs a live test.
- The groups come from past buying and will shift as habits change.

What we'd do next

- Run a 2 to 4 week A/B test of the product page box against best sellers.
- Try a simple bought or not bought input. It scored about 3 points higher in a side test (0.42 against 0.39).
- Refresh the model every night and look at the hit rate once a month.

No customer IDs appear in this memo. The method, code and every number are in the Git repository (see the README).